

HandsOn-3

Branching:

1. Create a new branch "**GitNewBranch**".
2. List all the local and remote branches available in the current trunk. Observe the "*" mark which denote the current pointing branch.
3. Switch to the newly created branch. Add some files to it with some contents.
4. Commit the changes to the branch.
5. Check the status with "**git status**" command.

Merging:

1. Switch to the master
2. List out all the differences between trunk and branch. These provide the differences in command line interface.
3. List out all the visual differences between master and branch using **P4Merge tool**.
4. Merge the source branch to the trunk.
5. Observe the logging after merging using "**git log --oneline --graph --decorate**"
6. Delete the branch after merging with the trunk and observe the git status.

Commands to add in created Repo

Branching Section

git branch GitNewBranch

- ◇ Creates a new branch named GitNewBranch from the current branch (master).

git branch -a

- ◇ Lists all **local** and **remote** branches.
 - The * marks the branch you are currently on.

git checkout GitNewBranch

- ◇ Switches your working directory from master to GitNewBranch so any new changes/commits go into this branch.

```
echo "file content" > file1.txt
```

- ◇ Creates a new file named file1.txt and writes "file content" into it.

```
echo "another file" > file2.txt
```

- ◇ Creates a new file named file2.txt and writes "another file" into it.

```
git add file1.txt file2.txt
```

- ◇ Stages file1.txt and file2.txt so they will be included in the next commit.

```
git commit -m "Added file1.txt and file2.txt to GitNewBranch"
```

- ◇ Commits the staged files to the GitNewBranch branch with the given commit message.

```
git status
```

- ◇ Shows the current state of the working directory and staging area.
 - If you've committed everything, it will say working tree clean.
-

Merging Section

```
git checkout master
```

- ◇ Switches back from GitNewBranch to the master branch (target branch for the merge).

```
git diff master..GitNewBranch
```

- ◇ Shows text-based differences between the master branch and GitNewBranch — what would change if merged.

```
git difftool master..GitNewBranch
```

- ◇ Opens the same differences in a **visual diff tool** (if configured, e.g., P4Merge, VS Code).

```
git merge GitNewBranch
```

- ◇ Merges commits from GitNewBranch into master.
 - If no conflicts, it will fast-forward or create a merge commit.

```
git log --oneline --graph --decorate
```

- ◇ Shows a compact, graphical commit history with branch and tag decorations, useful to see the merge structure.

git branch -d GitNewBranch

- ◇ Deletes the local branch GitNewBranch (only if it has been fully merged into master).

git status

- ◇ Shows the final status after merging and branch deletion.
 - Should show you are on master and working tree clean.

git push origin master

- ◇ Pushes the updated master branch (with merged changes) to the origin remote (your GitHub repo) so the changes appear online.

Output Screens:

```
MINGW64:/c/Users/LAVANYA/GitDemo
LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git checkout master
Already on 'master'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git branch GitNewBranch

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git branch -a
* master
  remotes/origin/master

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git checkout GitNewBranch
Switched to branch 'GitNewBranch'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitNewBranch)
$ echo "file content" > file1.txt

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitNewBranch)
$ echo "another file" > file2.txt

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitNewBranch)
$ git add file1.txt file2.txt
warning: in the working copy of 'file1.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'file2.txt', LF will be replaced by CRLF the next time Git touches it

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitNewBranch)
$ git commit -m "Added file1.txt and file2.txt to GitNewBranch"
On branch GitNewBranch
nothing to commit, working tree clean

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitNewBranch)
$ git status
On branch GitNewBranch
nothing to commit, working tree clean
```

```
LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (GitNewBranch)
$ git checkout master
Switched to branch 'master'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git diff master..GitNewBranch

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git difftool master..GitNewBranch

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git merge GitNewBranch
Already up to date.

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git log --online --graph --decorate
fatal: unrecognized argument: --online

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git log --oneline --graph --decorate
* 8686a5d (HEAD -> master, origin/master, GitNewBranch) Added file1.txt and file
2.txt to GitNewBranch
* e5af4ff Add .gitignore to ignore *.log files and log/ directories
* b4d7583 Added welcome.txt file Contains welcome message for Git training

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git branch -d GitNewBranch
Deleted branch GitNewBranch (was 8686a5d).

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean
```

```
LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git push origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 387 bytes | 387.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Lavanya-Yannam/GitDemo.git
8d774af..1f7d96a master -> master
```

github.com/Lavanya-Yannam/GitDemo

Lavanya-Yannam / GitDemo

Q Type to sea

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

GitDemo

Public

Pin

Watch 0

master 1 Branch 0 Tags

Go to file

Add file

<> Code

Lavanya-Yannam

Adding file1.txt and file2.txt

1f7d96a · 1 minute ago

6 Commits

.gitignore	Add .gitignore to ignore *.log files and log/ directories	2 hours ago
file1.txt	Adding file1.txt and file2.txt	1 minute ago
file2.txt	Adding file1.txt and file2.txt	1 minute ago
welcome.txt	Added welcome.txt file	4 hours ago

README

Add a README

Help people interested in this repository understand your project by adding a README.

Add a README

Lavanya-Yannam / GitDemo

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Files

master +

Go to file

.gitignore

file1.txt

file2.txt

welcome.txt

GitDemo / file1.txt

Lavanya-Yannam

Adding file1.txt and file2.txt

Code

Blame

1 lines (1 loc) · 13 Bytes

1

file content

